Chapter 8. Deflection of beam

8.1. Introduction

Couples and transverse forces applied to beams cause them to deflect laterally. In Chapter 9 we determined a relationship between the curvature of the deflection curve and the bending moment at a cross section.

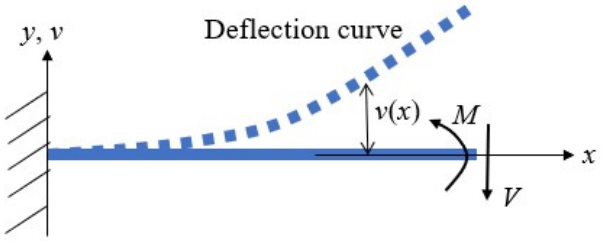
$$\frac{1}{\rho} = \frac{M}{EI} \qquad (9.5)$$

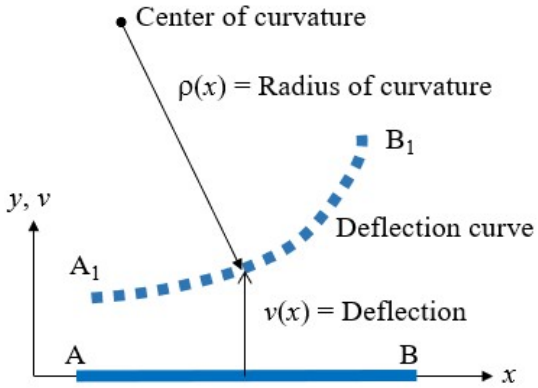
In which:

ρ - Radius of curvature of the bent neutral plane of the beam.

We define x as the axis of beam, v as the displacement of the beam in the y direction of any point on the axis of the beam. Because the y axis is positive upward, the deflections are also positive when upward.

We can represent the deflection of a beam by giving v as a function of x. Such a function is called ***deflection curve***.





AB – original beam.

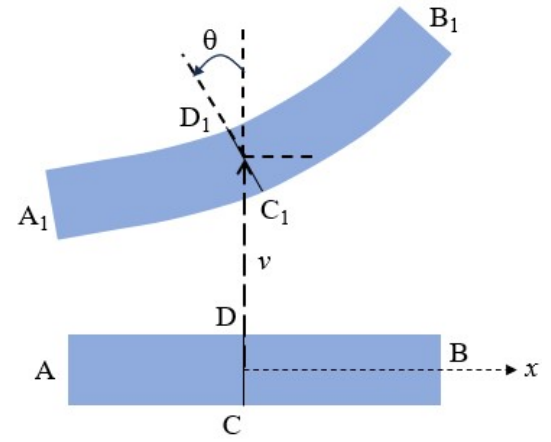
A_1B_1 – deflected beam.

8.2. Equation of the deflection curve

Because of bending, the cross-section of the beam not only deflect, but also make a rotation (see right figure, the cross-section CD deflects/rotates to C_1D_1). The angle of rotation of the cross-section is denoted as θ. The slope $(\frac{dv}{dx})$ is equal to $\tan\theta$. That is, $\tan\theta = \frac{dv}{dx}$. Since the rotation is small compared with unity, then $\theta = \tan\theta$ and the approximation

$$\theta = \frac{dv}{dx} \qquad \text{(a)}$$

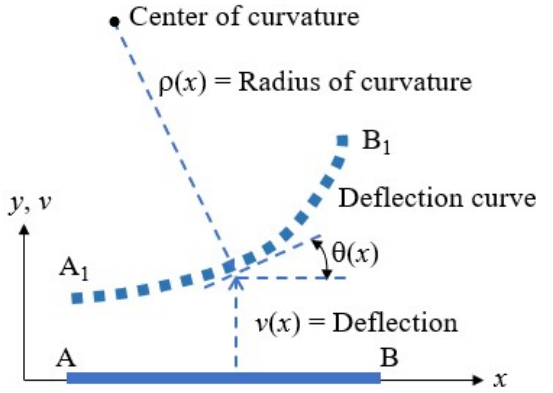
can be used to find θ. θ can also be understood as the rotation of the beam when bending.



Both deflection v and the rotation θ are in the x - y plane.

From calculus, the radius of curvature is related to the derivatives of the deflection curve by

$$\frac{1}{\rho} = \frac{\frac{d^2v}{dx^2}}{\left[1 + \left(\frac{dv}{dx}\right)^2\right]^{3/2}}$$



As the slope $(\frac{dv}{dx})$ of the deflected beam is small compared with unity, the approximation

$$\frac{1}{\rho} = \frac{d^2v}{dx^2} \quad \text{(b)}$$

can be treated as equality. We have now achieved our aim of finding a relationship between deflection v and the radius of curvature ρ.

To obtain the relationship between v and M, We substitute Eq. (9.5) into Eq. (a),

$$\frac{d^2v}{dx^2} = \frac{M}{EI} \quad (8.1)$$

Equation (8.1) is still not an expression for the deflection curve of the beam. We will refer to this equation as the *moment-curvature equation*. It is a second-order, ordinary differential equation that we can integrate to determine the slope and deflection, given an expression for the bending moment (M) and appropriate boundary conditions and continuity conditions.

8.3. Slope and deflection by integration

The deflection and slope can be solved by integration of the second-order differential, Eq. (8.1).

All solutions follow a straightforward procedure. We will outline that procedure.

Integration of the second-order differential equation – *second-order method*

Once we know the equation of the bending moment, we are able to solve the differential equation (8.1) for v. This involves integrating twice which will produce two constants of integration.

For a general form of the moment M, integrating the equation (8.1) gives an expression for the slope of the beam:

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{EI}\int M\mathrm{d}x + C_1 \qquad (8.2)$$

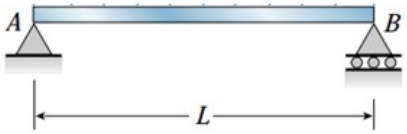
Integrating again gives an expression for the deflection of the beam:

$$v = \frac{1}{EI}\int\int M\mathrm{d}x\mathrm{d}x + C_1x + C_2 \qquad (8.3)$$

The constants C_1 and C_2 are to be determined from the boundary conditions of the beam (the beam's supports).

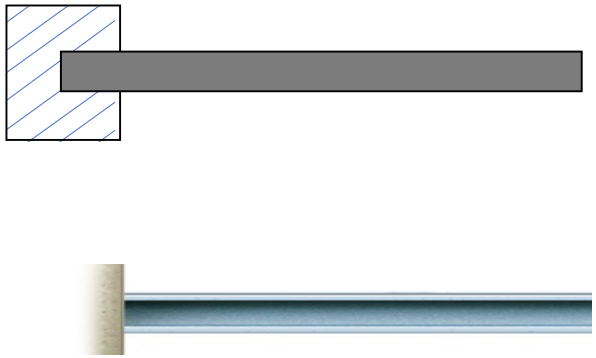
8.4. Boundary conditions (BCs)

The load-deflection is to be solved under specified boundary conditions. The most common supports are pin/roller and fix end.



Simply supported end (pin - roller)

$v = 0, \qquad M = 0$



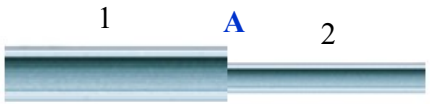
Clamped (fixed) end

$v = 0, \theta = 0$ or $v = 0, \quad \frac{dv}{dx} = 0$

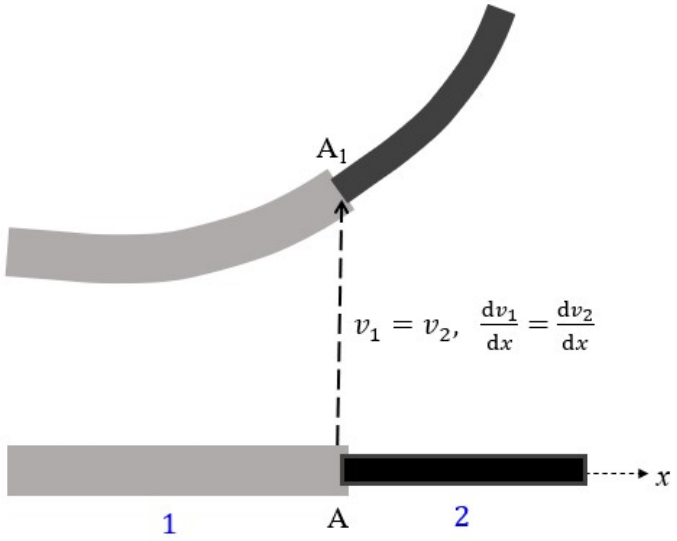
8.5. Continuity conditions (CCs)

More complicated beams require application of requirement for continuity of displacement, slope, shear force and bending moment.

Ideal connection

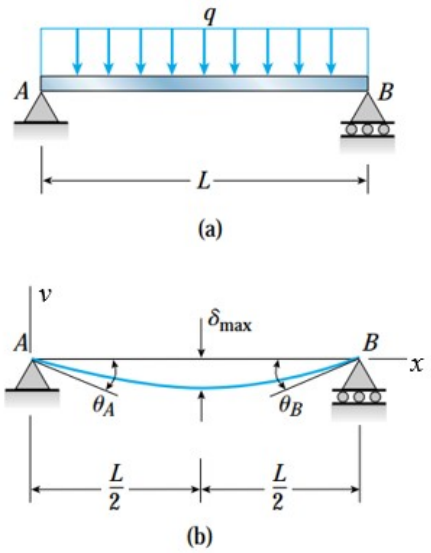


At A, $v_1 = v_2, \frac{dv_1}{dx} = \frac{dv_2}{dx}$



Subscripts 1 and 2 denote the beams 1 and 2, respectively.

Example 1. Find the deflection curve and the maximum deflection of a simple beam with a uniform load shown in Fig. a. (*Note*: The beam has length L and constant flexural rigidity EI.)



$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution:

Since we will be analyzing this beam by solving the bending-moment equation, we must begin with a general expression for the moment.

Using a free-body diagram (Chapter 8), the internal bending moment of the beam can be obtained as

$$M = \frac{q}{2}(-x^2 + Lx) \qquad \text{(a)}$$

Putting Eq. (a) into Eq. (8.2) and integrating, one obtains the slope function of the beam:

$$\frac{dv}{dx} = \frac{q}{12EI}(-2x^3 + 3Lx^2) + C_1 \qquad \text{(b)}$$

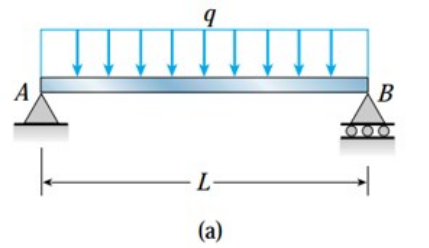
Integrating Eq. (b) yields the deflection function of the beam:

$$v = \frac{q}{24EI}(-x^4 + 2Lx^3) + C_1 x + C_2 \qquad \text{(c)}$$

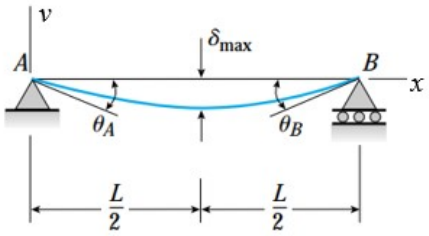
The constants C_1 and C_2 can be determined from the boundary conditions. As shown in the deflection diagram of Fig. b, the deflections at the left and right ends of the beam are zero:

(1) At $x = 0$: $v = 0$; So, $C_2 = 0$

(2) At $x = L$: $v = 0$; So, $C_1 = -\frac{qL^3}{24EI}$



(a)



(b)

Substituting the constants C_1 and C_2 into Eqs. (b) and (c), we get the slope and deflection functions of the beam:

$$\frac{dv}{dx} = \frac{q}{24EI}(-4x^3 + 6Lx^2 - L^3) \qquad \text{(d)}$$

$$v = \frac{q}{24EI}(-x^4 + 2Lx^3 - L^3x) \qquad \text{(e)}$$

The maximum deflection of the beam is at the midpoint ($x = L/2$):

$$\delta_{\max} = v\Big|_{x=L/2} = -\frac{5qL^4}{384EI} \qquad \text{(f)}$$

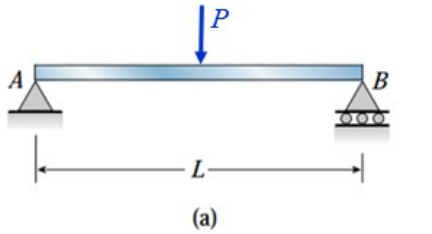
The negative sign represents that the deflection is downward.

The maximum slope of the beam is at the left and right ends ($x = 0$ and $x = L$). For example, the rotation at the left end A is

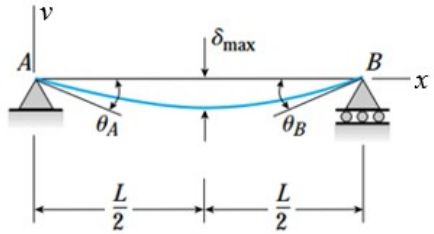
$$\theta_A = \frac{dv}{dx}\Big|_{x=0} = -\frac{qL^3}{24EI} \qquad \text{(g)}$$

The negative sign represents that the rotation θ_A is clockwise.

Example 2. Find the deflection curve and the maximum deflection of a simple beam with a concentrated load applied at the midpoint of the beam shown in Fig. a.



(a)



(b)

$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

Due to symmetry, we can consider the left part of the beam only.

Using a free-body diagram (Chapter 8), the internal bending moment of the beam can be obtained as

$$M = \frac{Px}{2} \quad , \quad x \le \frac{L}{2} \qquad \text{(a)}$$

Substitute Eq. (a) into Eqs. (8.2) and (8.3), and integrate, one obtains:

$$\frac{dv}{dx} = \frac{P}{4EI}x^2 + C_1, \quad x \le \frac{L}{2} \qquad \text{(b)}$$

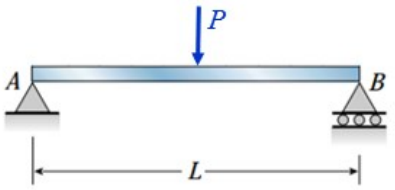
Integrate Eq. (b), one obtains the deflection function of the beam:

$$v = \frac{P}{12EI}x^3 + C_1 x + C_2, \quad x \le \frac{L}{2} \qquad \text{(c)}$$

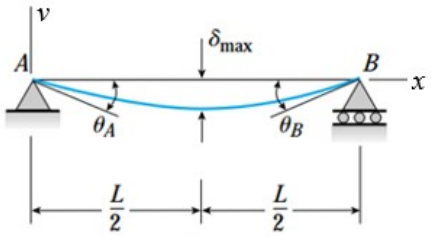
The constants C_1 and C_2 should be determined from the boundary conditions. As illustrated in the deflection diagram of Fig. b, the deflections at both ends of the beam are zero. Because the deformations of the left and the right parts of the beam are mirror symmetric, the slope at the midpoint of the beam is also zero:

(1) at $x = 0$: $v = 0$; So, $C_2 = 0$

(2) at $x = L/2$: $dv/dx = 0$; So, $C_1 = -\frac{PL^2}{16EI}$



(a)



(b)

Substituting the constants C_1 and C_2 into Eq. (b) and (c), we get the slope function and deflection curve of the beam as

$$\frac{dv}{dx} = \frac{P}{16EI}(4x^2 - L^2), \quad x \le \frac{L}{2} \qquad \text{(d)}$$

$$v = \frac{P}{48EI}(4x^3 - 3L^2x) \quad , \quad x \le \frac{L}{2} \qquad \text{(e)}$$

The maximum deflection is at the midpoint of the beam:

$$\delta_{max} = v\Big|_{x=L/2} = -\frac{PL^3}{48EI} \qquad \text{(f)}$$

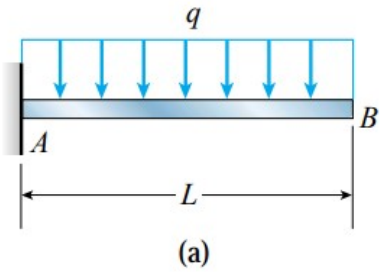
The negative sign represents that the deflection is downward.

The maximum slope of the beam is at the left and right ends ($x = 0$ and $x = L$). For example, θ_A is

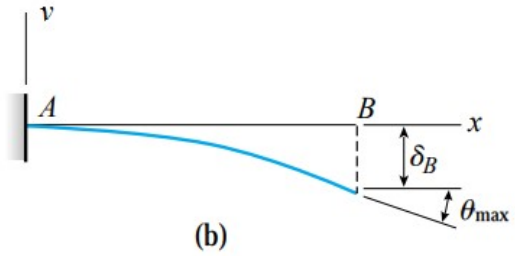
$$\theta_A = \frac{dv}{dx}\Big|_{x=0} = -\frac{PL^2}{16EI} \qquad \text{(g)}$$

The negative sign represents that the rotation θ_A is clockwise.

Example 3. Find the deflection curve and the maximum deflection of the cantilever beam with a uniform load shown in Fig. a.



(a)



(b)

$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C_1 \qquad (8.2)$$

Solution

Using a free-body diagram (Chapter 8), the internal bending moment of the beam can be obtained as

$$M = -\frac{q}{2}(x^2 - 2Lx + L^2) \qquad \text{(a)}$$

Since the bending moment is given, one can integrate Eq. (8.2) to get the slope function of the beam as

$$\frac{dv}{dx} = \frac{q}{6EI}(-x^3 + 3Lx^2 - 3L^2x) + C_1 \qquad \text{(b)}$$

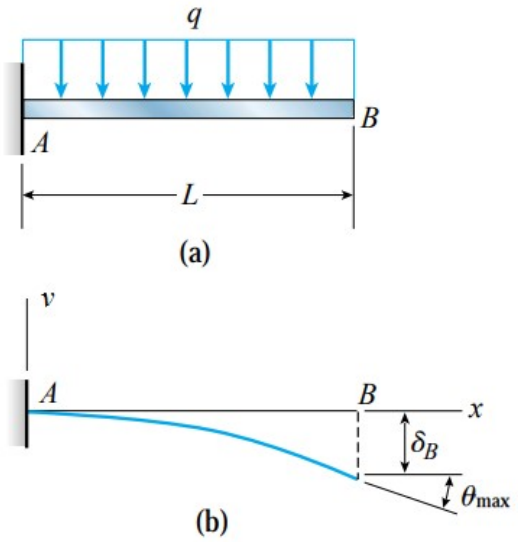
Equation (b) can be integrated to give the deflection function of the beam:

$$v = \frac{q}{24EI}(-x^4 + 4Lx^3 - 6L^2x^2) + C_1x + C_2 \qquad \text{(c)}$$

The load tends to deflect the beam as shown in Fig. b. Since the left end of the beam is fixed, the deflection and slope at point A are zero. These yield:

at $x = 0$: $v = 0$ and $dv/dx = 0$

So, $C_1 = 0$ and $C_2 = 0$



Substituting the constants C_1 and C_2 into Eqs. (b) and (c), we can get the slope function and deflection curve of the beam as

$$\frac{dv}{dx} = \frac{q}{6EI}(-x^3 + 3Lx^2 - 3L^2x) \qquad \text{(d)}$$

$$v = \frac{q}{24EI}(-x^4 + 4Lx^3 - 6L^2x^2) \qquad \text{(e)}$$

The maximum deflection is at the free end of the beam, $x = L$:

$$\delta_{max} = v\Big|_{x=L} = -\frac{qL^4}{8EI} \qquad \text{(f)}$$

The rotation of the beam is equal the slope. At the free end $x = L$, the rotation of the beam is

$$\theta_{max} = \frac{dv}{dx}\Big|_{x=L} = -\frac{qL^3}{6EI} \qquad \text{(g)}$$

The directions of the deflection and rotation of the beam is shown in Fig. b.